

SIDDHARTH TRIPATHI

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PROFILE

Computer Science Engineering student with hands-on expertise in deep learning, computer vision, and IoT systems. Specialized in building production-ready ML models, neural network optimization, and full-stack AI applications. Passionate about physical AI, robotics, and edge deployment of intelligent systems.

EDUCATION

SRM Institute of Science and Technology

B.Tech in Computer Science Engineering | 2025 – Present (2nd Year)

Relevant Coursework: Software & Systems Engineering, Mathematical & Algorithmic Concepts, Artificial Intelligence & Machine Learning

TECHNICAL EXPERTISE

Deep Learning & Neural Networks: PyTorch, CNN, ResNet, Transfer Learning, Fine-tuning, Backpropagation, Hyperparameter Tuning, Model Optimization, Quantization, Pruning

Computer Vision & Image Processing: OpenCV, Real-time Inference, Image Classification, Object Detection, Transfer Learning, Edge Deployment, Inference Optimization

Machine Learning & Data Science: Scikit-learn, NumPy, Pandas, Evaluation Metrics, Cross-validation, Feature Engineering, Data Preprocessing, Model Evaluation

Full-Stack Development: Python, React, JavaScript, HTML, CSS, Streamlit, API Integration, REST APIs, Web Applications

Tools & Platforms: Git, GitHub, VS Code, Google Colab, Kaggle, LLM APIs (Google Gemini), PyTorch Lightning, Jupyter Notebooks

SKILLS

Languages: C/C++, Java, Python

Web Technologies: React, HTML, CSS, JavaScript

ML/AI Libraries & Frameworks: NumPy, Pandas, Scikit-learn, PyTorch, Torchvision, OpenCV, Streamlit, StreamlitWebRTC, Matplotlib, C++ STL

Core Competencies: Deep Learning, Neural Networks, Computer Vision, Transfer Learning, Model Optimization, Inference, Edge Computing, Artificial Intelligence, Data Structures & Algorithms, Embedded Systems, Software Engineering

PROJECTS

Breast Cancer Classification – Custom Neural Network Implementation

- Engineered a fully connected neural network from scratch using NumPy, implementing forward propagation, backpropagation, ReLU/sigmoid activation functions, cross-entropy loss computation, and stochastic gradient descent (SGD) optimization with learning rate scheduling. Demonstrated deep understanding of fundamental neural network mechanics without relying on high-level deep learning frameworks.
- Trained and evaluated neural network on Wisconsin Breast Cancer Dataset (569 samples, 30 features), achieving 95%+ classification accuracy and strong F1-scores, validating correctness of gradient descent optimization and backpropagation implementation.

Generative AI Study Assistant – Document Query Platform with LLM Integration

- Problem: Students spend 30+ minutes re-reading study materials to find specific information. Solution: Built an AI-powered study assistant that leverages LLM reasoning (Google Gemini API) for instant, context-aware explanations.
- Engineered full-stack application using Python and Streamlit with interactive document upload and query interface, enabling users to retrieve instant, AI-generated explanations from study materials without manual searching. Integrated Google Gemini LLM API for semantic understanding and context-aware response generation.
- Impact: Reduced study lookup time by 80%, processed 500+ user queries during beta testing, validating product-market fit and demand for AI-assisted learning.

EcoSort AI – Real-Time Waste Classification System with Edge Deployment

- Engineered a real-time waste classification deep learning model using PyTorch and ResNet18 architecture via transfer learning, achieving 94.2% top-1 accuracy on 6-category waste classification task (cardboard, glass, metal, paper, plastic, trash) trained on 12,000+ labeled waste images.
- Optimized neural network inference pipeline through quantization and pruning techniques, reducing model latency from 450ms → 120ms per inference, enabling real-time processing at 30 FPS on resource-constrained edge devices. Reduced model size by 35% through post-training quantization.
- Developed full-stack web application (Python, Streamlit, OpenCV, Streamlit-WebRTC) with live webcam inference, confidence scoring, and color-coded visual feedback. End-to-end latency optimized to <150ms, supporting real-time classification at 30 FPS for both webcam and image-based inference.
- Implemented data augmentation and transfer learning strategies, reducing GPU training time from 12 hours (training from scratch) to 4.8 hours, demonstrating 60% reduction in training time through fine-tuning approach on pre-trained ResNet18 backbone.

EXPERIENCE & ACHIEVEMENTS

HACKSAGON – National-Level Hackathon, Top 5 Finalist

- Ranked 5th out of 150+ competing teams across 8 states in a highly competitive 24-hour national hackathon, finishing directly behind the track winner. Demonstrated ability to ideate, develop, and pitch a working technical solution under extreme time constraints.
- Collaborated cross-functionally with team members to build an AI-powered application from concept to working prototype in <24 hours, integrating machine learning inference with web technologies, and presented solution to a panel of industry expert judges under competitive evaluation criteria.
- Demonstrated leadership in problem-solving, technical depth in ML deployment, and communication skills through high-pressure pitch and live demonstration of working software to competition judges.